

Hankel-Norm and Information Limits on Delay Detection by Low-Order State-Space Models, with Application to Graph Random Walks

Olivia Kim and Aryan Padarthy

Abstract—An order- S linear time-invariant (LTI) state-space model must decide whether the current sample repeats the one k steps earlier, as random-walk graph learners do to detect cycles. For Gaussian inputs, exact detection requires $S \geq k + 1$ and a singular state matrix, so LTI invertible dynamics such as S4D (or Mamba with a fixed step) cannot achieve it. Since the lag- k delay has k unit Hankel singular values, for $S < k$ no linear probe beats the zero probe in Hankel norm, any probe explains at most S/k of the delayed sample's variance, and every readout has detection divergence at most $\frac{1}{2} \ln \frac{k}{k-S}$. Optimal fits track the S/k law, and a trained shift register switches on at $S = k + 1$. Trained diagonal SSMs need $4.0\times$ the minimal state to reach AUROC 0.95; frozen optimal poles remove most of this overhead for $k \leq 12$. On circular skip-link graphs the remaining gap lies downstream of the dynamics.

Index Terms—State-space models, Hankel operators, model order reduction, detection, random walks on graphs.

I. INTRODUCTION

DECIDING whether a signal repeats itself after a lag k is a basic detection task: it is echo detection, and it is the primitive that sequence models must implement to recall earlier inputs. Structured state-space models (SSMs) [1]–[3] implement it with a recursive filter of order S , and much of their design is about approximating delays with few states [4], [5]. That the pure delay z^{-k} has McMillan degree k is classical realisation theory [6]–[8]. What is missing is a *rate*: how much of the delayed sample can an order- S state carry when $S < k$, and what does that imply for detection by an arbitrary nonlinear readout? Related limits for recurrent models concern memory decay and optimisation [9], copying and associative recall [10], [11], and state tracking [12].

Our application is graph learning. Message-passing GNNs are bounded by the 1-WL test [13], [14] and cannot separate the regular circular skip-link (CSL) graphs [15], [16]. Walk-based learners read random walks as sequences [17]–[19], and random node features [20], [21] turn walk revisits, and hence closed walks, into token repetitions. SSMs are now used as the sequence backbone of such models [18], [22], [23].

Our contributions are: (i) *exact limits*: for Gaussian inputs, exact lag- k detection needs $S \geq k + 1$ and a singular state matrix, and is impossible for invertible dynamics at any S ;

(ii) *rates*: for $S < k$, no linear probe beats the zero probe in Hankel norm, the explained variance is at most S/k , and for Gaussian inputs this bounds the divergence available to every readout; (iii) *graph consequence*: exact non-backtracking (NB) return horizons $W^*(r)$ at which window features separate each CSL class; and (iv) *validation*: optimal fits, shift-register and frozen-pole controls, and seeded experiments that separate the information limit, the optimisation of the dynamics and the readout.

II. PROBLEM SETTING

A sequence model with real state $s_t \in \mathbb{R}^S$ reads tokens $x_t \in \mathbb{R}^d$:

$$s_t = A_t s_{t-1} + B_t x_t, \quad \hat{y}_t = g(s_t, x_t), \quad (1)$$

with an arbitrary readout g . It is *LTI* if $A_t \equiv A$ and $B_t \equiv B$. The *lag- k repeat* label is $y_t = 1[x_t = x_{t-k}]$. In the graph setting, $v_0, v_1, \dots$ is an NB walk ($v_{t+1} \neq v_{t-1}$), each walk draws fresh features $f_v \sim \mathcal{N}(0, I_d)$ i.i.d. over nodes, and $x_t = f_{v_t}$. Then $y_t = 1[v_t = v_{t-k}]$ almost surely, i.e. $y_t = 1$ exactly when the last k steps form a closed NB walk. Their frequencies follow from powers of the Hashimoto (NB) operator [24], [25].

III. LIMITS

Proposition 1 (Exact detection). *Let x_t be i.i.d. $\mathcal{N}(0, I_d)$ with a stationary past, and test $H_1: x_t = x_{t-k}$ against $H_0: x_t$ fresh, from (s_t, x_t) of a stable LTI model. (a) If $S \leq k$, or (b) if A is invertible (any S), every readout g has Bayes error > 0 . (c) If $S \geq k + 1$, a singular A suffices: the shift register $s_t = (w^\top x_t, \dots, w^\top x_{t-S+1})$ with $g = 1[w^\top x_t = (s_t)_{k+1}]$ has zero error for every $w \neq 0$.*

Proof: Under both hypotheses (s_t, x_t) is zero-mean Gaussian, so the two laws are either equivalent (error > 0) or mutually singular. The support under H_1 lies in that under H_0 , where x_t is free given s_{t-1} . So singularity needs $a^\top s_t + b^\top x_t = 0$ a.s. under H_1 but not under H_0 . Since s_{t-1} has the same law under both, this means $a^\top A s_{t-1} = v^\top x_{t-k}$ a.s. with $v = -(B^\top a + b) \neq 0$, i.e. $a^\top A^m B = \delta_{m,k} v^\top$ for all $m \geq 1$: an exact FIR realisation of $v^\top z^{-k}$ by (A, AB, a) . (a) Its Hankel matrix has rank k , so $S \geq k$. At $S = k$ the realisation is minimal, hence nilpotent and controllable through AB ; but $[AB, \dots, A^k B] = A[B, \dots, A^{k-1} B]$ has rank below k for nilpotent A . (b) If A is invertible, $a^\top A^{k+1} B = 0$ for all $j \geq 0$ makes a orthogonal to $A^{k+1} \mathcal{R} = \mathcal{R}$, the reachable space, so $v = 0$. (c) Under H_0 , $w^\top x_t \neq w^\top x_{t-k}$ a.s. ■

Manuscript received XX; revised XX. The associate editor coordinating the review of this manuscript and approving it for publication was XX.

The authors are with the Allen Independent School District, Allen, TX, USA (e-mail: olivia.kim@student.allenisd.org; aryan.padarthy@student.allenisd.org).

Code and data: <https://github.com/The-MathKing/ITE>.

At $S = k$ the error in (a) can nonetheless be made arbitrarily small (Theorem 1), so the threshold $S \geq k + 1$ concerns exactness, not a practical barrier.

Exactness over *all* token sequences is stronger: it needs $S \geq (k + 1)d$, because the subspaces $A^j B \mathbb{R}^d$, $j = 0, \dots, k$, must be independent.

Every S4D discretisation $A = \exp(\Delta\Lambda)$, and Mamba's with a fixed step, is invertible, so exact detection is never available to these LTI models. The relevant question is how *well* an order- S state can carry the lag- k sample. A linear probe $z_t = c^\top s_t + \beta^\top x_t$ estimates $w^\top x_{t-k}$ with $\|w\| = 1$. Its error $z_t - w^\top x_{t-k} = \sum_{m \geq 0} e_m x_{t-m}$ has taps $e_m = c^\top A^m B - \delta_{m,k} w^\top$ for $m \geq 1$ and transfer function $E(z) = \sum_{m \geq 1} e_m z^{-m}$, with $\|E(e^{j\omega})\|$ the Euclidean norm of the $1 \times d$ row. Let $\mathcal{H}_k(f)$ be the $k \times kd$ block-Hankel section with (i, j) block f_{i+j+1} , $0 \leq i, j < k$.

Theorem 1 (Hankel limits). *If $S < k$, every LTI probe satisfies*

$$\|\mathcal{H}_k(e)\|_2 \geq 1, \quad \text{hence} \quad \sup_\omega \|E(e^{j\omega})\| \geq 1, \quad (2)$$

$$\|\mathcal{H}_k(e)\|_F^2 = \sum_{m=1}^{2k-1} \min(m, 2k-m) \|e_m\|^2 \geq k - S. \quad (3)$$

Hence, for i.i.d. tokens with $\mathbb{E}[xx^\top] = I_d$ and a stationary past (or $t \geq 2k-1$), $\mathbb{E}[(z_t - w^\top x_{t-k})^2] \geq 1 - S/k$: the probe explains at most a fraction S/k of the variance. For $S = k$ the infimum of the error is 0, but it is not attained.

Proof: Since $h_{i+j+1} = (c^\top A^i)(A^{j+1}B)$, we have $\text{rank } \mathcal{H}_k(h) \leq S$. The target $\delta_{m,k} w^\top$ gives a block anti-diagonal matrix with k unit singular values. The Eckart–Young–Mirsky theorem [26], [27] in the spectral and Frobenius norms gives $\|\mathcal{H}_k(e)\|_2 \geq \sigma_{S+1} = 1$ and $\|\mathcal{H}_k(e)\|_F^2 \geq \sum_{i>S} \sigma_i^2 = k - S$. A finite section is bounded by the Hankel operator Γ_E , and $\|\Gamma_E\| \leq \|E\|_\infty$ because Γ_E is a compression of the Laurent operator of E . This gives (2); Nehari's theorem [28] sharpens it to $\|\Gamma_E\| = \text{dist}(E, H^\infty)$. The tap e_m occurs $\min(m, 2k-m) \leq k$ times, which gives (3) and the variance bound. For $S = k$, take $A = \gamma P$ with P the k -cycle permutation, $B = u_1 w^\top$, $c = \gamma^{-k} u_1$ and $\beta = -\gamma^{-k} w$. Then $e_m = \gamma^{m-k} w^\top$ for $m \in \{2k, 3k, \dots\}$ and 0 otherwise, so the error is $\gamma^{2k}/(1 - \gamma^{2k}) \rightarrow 0$ as $\gamma \rightarrow 0$. Zero error would be an exact FIR realisation, which the proof of Proposition 1(a) excludes at $S = k$. ■

Bound (2) says that for $S < k$ no LTI probe beats outputting zero in Hankel norm: all Hankel singular values of z^{-k} equal one, so its optimal Hankel-norm approximant of degree $< k$ is zero [7], [8]. Bound (3) is the rate. Under Gaussian inputs it extends to every readout.

Corollary 1 (Any readout). *Let x_t be i.i.d. $\mathcal{N}(0, I_d)$ and $S < k$. For every measurable g , the minimum mean-squared error of x_{t-k} given (s_t, x_t) is at least $d - S/k$. For H_1 : $x_t = x_{t-k}$ against H_0 : x_t fresh, the observations satisfy $\text{KL}(P_1 \| P_0) \leq \frac{1}{2} \ln \frac{k}{k-S}$, so no detector exceeds balanced accuracy $\frac{1}{2} + \frac{1}{2} \left(\frac{1}{4} \ln \frac{k}{k-S} \right)^{1/2}$.*

Proof: For jointly Gaussian variables the conditional mean is linear. Stacking d probes gives a target with kd unit Hankel singular values, and (3) yields total error $\geq d - S/k$.

Next, (s_t, x_t) is a function of (s_{t-1}, x_t) . These are independent under H_0 and have the same marginals under H_1 , so $\text{KL} = I(s_{t-1}; x_{t-k}) = -\frac{1}{2} \sum_i \ln(1 - \lambda_i)$, where λ_i are the squared canonical correlations. Relative to s_{t-1} , x_{t-k} is the lag- $(k-1)$ tap, with the $m = 0$ tap included. The $k \times k$ section with blocks $c^\top A^{i+j} B$, $0 \leq i, j < k$, again has a target with k unit singular values, and tap m occurs $\min(m+1, 2k-1-m) \leq k$ times. Hence $\sum_i \lambda_i \leq S/k$. By convexity the maximum puts all mass on one λ_i . Data processing and Pinsker's inequality complete the proof. ■

Remark 1. The bounds limit *precision*, not information. For $S < k$ the state can still correlate with the lagged sample, so statistics pooled over many tokens remain informative. The near-optimal realisation at $S = k$ needs gains γ^{-k} and is exponentially ill-conditioned (cf. [9]). Input-dependent (A_t, B_t) break the factorisation used above, so we test them empirically. On walks, revisits make tokens dependent. Eq. (2) is distribution-free; the variance and divergence rates use the i.i.d. surrogate, which we check in Section IV.

NB return horizons. Let $p_G(\ell) = \Pr[v_\ell = v_0]$ for the stationary NB walk (uniform on directed edges). We compute it exactly from the Hashimoto operator [24]. For a family $\{G_r\}$, let $W^*(r)$ be the smallest W such that $\max_{\ell \leq W} |p_{G_r}(\ell) - p_{G_{r'}}(\ell)| > \tau$ for all $r' \neq r$. It is the shortest window of revisit frequencies that separates class r . For the ten CSL classes ($n = 41$, skips $r \in \{2, 3, 4, 5, 6, 9, 11, 12, 13, 16\}$), $W^*(r)$ ranges from 3 to 8 and is unchanged for $\tau \in [10^{-4}, 3 \cdot 10^{-3}]$ (Table I). W^* is a horizon for *exact* access. An order- S reader with $S < W^*$ still explains up to S/ℓ of the lag- ℓ variance. We therefore expect pooled estimators to succeed below W^* , at a sample cost that grows as S/W^* shrinks, since each token carries divergence at most about $S/(2\ell)$.

IV. EXPERIMENTS

Models. Our learned SSM is one diagonal complex layer with S4D-Lin initialisation [2]. It has $\lfloor S/2 \rfloor$ complex modes plus one real mode if S is odd, so its real state dimension is exactly S . The dynamics are $A = \exp(\Delta\Lambda)$ and $B_t = \Delta B$, with Δ learned per mode (LTI) or set to $\Delta_t = \text{softplus}(W_\Delta x_t + v_\Delta)$ (input-dependent step, the Mamba-style gate). The readout sees $[\Re s_t, \Im s_t, x_t]$ and comparison features $(Ux_t - Vs_t)^{\odot 2}$, followed by a two-layer MLP. We use $d = 8$ features, AdamW with a cosine schedule, and 3 seeds; intervals are 95% t -intervals. All runs are on CPU, and a script regenerates every number and figure in this letter.

Delay realisation. We fit an S -state diagonal impulse response to $\delta_{m,k}$ ($m \geq 1$), keeping the best of four restarts (Fig. 1). All 16 fits with $S < k$ respect $\|e\|_2 \geq \sqrt{1 - S/k}$, with a median excess MSE of 0.11 over the bound. The error first drops to about 0.1 at a median of $S = 1.0k$, so the rate of Theorem 1 is nearly attained by direct fitting.

Token-level detection. We train on NB walks ($T = 64$) over 256 random 4-regular graphs ($n = 24$, unions of two Hamiltonian cycles) and test on 64 unseen graphs. Fig. 2 shows that the test AUROC for y_t rises smoothly with S . There is no sharp transition at $S = k$, where the AUROC is 0.69–0.72. The state needed for a given AUROC grows

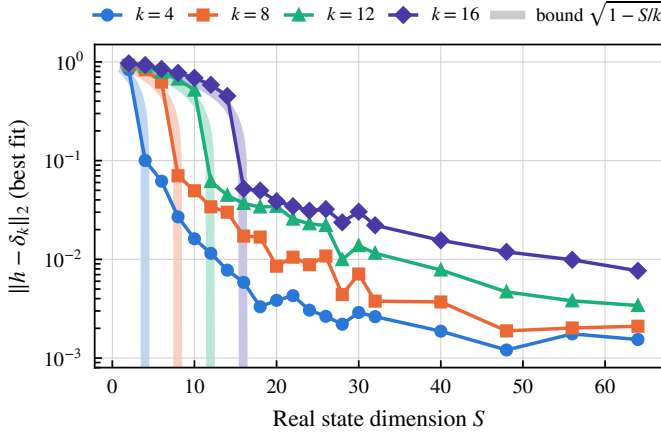


Fig. 1. Best gradient fit (four restarts, taps $m \leq 63$) of an S -state diagonal LTI impulse response to the lag- k delay. Shaded curves: the lower bound $\sqrt{1 - S/k}$ of Theorem 1 for $S < k$.

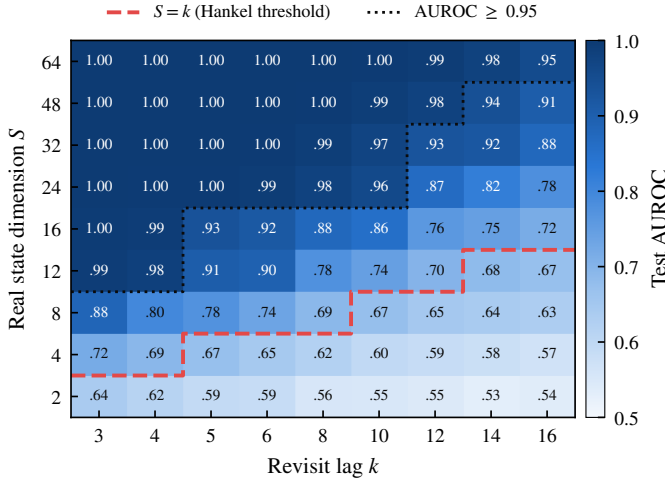


Fig. 2. Test AUROC of the LTI model for lag- k revisit detection on NB walks over random 4-regular graphs ($n = 24$, $T = 64$, $d = 8$; mean of 3 seeds). Red dashed line: $S = k$. Dotted line: first S with $\text{AUROC} \geq 0.95$.

approximately linearly in k . Its median ratio to k is 2.0, 2.7 and 4.0 at AUROC thresholds 0.8, 0.9 and 0.95; an AUROC of 0.99 is reached for only 7 of the 9 lags. Below the bound, all 22 LTI cells lie under the optimal AUROC of the Gaussian surrogate with explained variance S/k (Fig. 3). We checked the surrogate against walk structure with logistic regression on all pairwise coincidences among the S most recent tokens, which is what a shift register can compare. For $S \leq k - 3$ these predict lag- k revisits at AUROC 0.49–0.51, i.e. not at all. Within two lags of k they reach up to 0.53 ($S = k - 2$), 0.58 ($S = k - 1$) and 0.67 ($S = k$), because a closed walk that leaves and returns along the same path makes stored tokens coincide. The surrogate is therefore fair for $S \leq k - 3$. The input-dependent step changes the AUROC by -0.001 ± 0.002 (mean over cells, $|\Delta| \leq 0.03$; Wilcoxon $p = 0.005$, paired t -test $p = 0.14$), a negligible difference. Its learned Δ_t varies by only 1% across tokens, which is expected because random features give the gate nothing to select on.

Controls. Three controls separate the information limit from

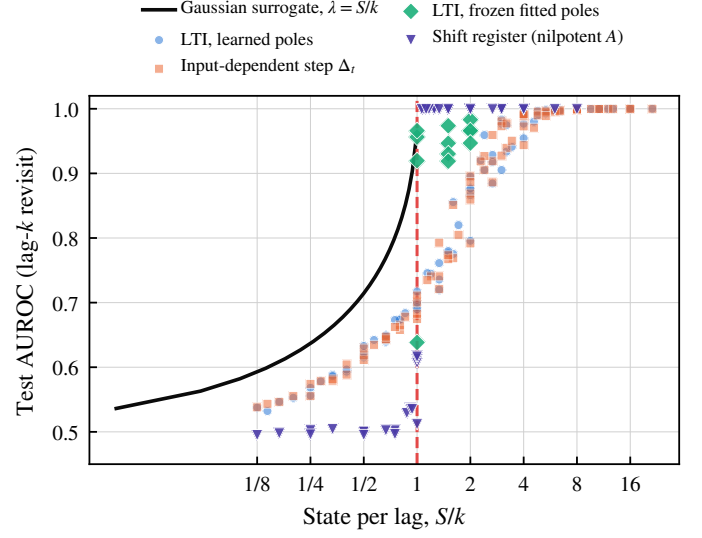


Fig. 3. Test AUROC against S/k (all k , seed means). Solid curve: optimal AUROC of the i.i.d. Gaussian surrogate with explained variance S/k (Corollary 1). Red dashed line: $S = k$.

training effects (Fig. 3).

- **Shift register.** A trained shift register (nilpotent A , learned w , $k \in \{4, 8, 12, 16\}$) has AUROC 0.51–0.62 at $S = k$ and 1.00 at $S = k + 1$. This is the switch predicted by Proposition 1(c). The excess over 0.5 at $S = k$ matches the pairwise coincidences above.
- **Frozen poles.** We freeze the poles at the delay fit of Fig. 1 and train only B and the readout. The AUROC is 0.87 at $S = k$ and 0.97 at $S = 2k$ (mean over k), against 0.70 and 0.85 with learned poles. For $k \leq 12$ it is already 0.92–0.97 at $S = k$. At $k = 16$ it drops to 0.64 ± 0.26 , where the fitted poles are ill-conditioned (Remark 1), and it reaches 0.95 only at $S = 2k$, half the state that learned poles need.
- **Comparison features.** Removing them changes the $k = 8$ AUROC by at most 0.02 at any S .

For $k \leq 12$ the information limit is therefore nearly attainable at $S \approx k$, and most of the overhead of trained SSMS comes from learning the dynamics, not from the readout. At $k = 16$ the conditioning of the optimal realisation accounts for about half of the overhead.

CSL classification. CSL graphs are 4-regular and featureless, so 1-WL is provably at chance (10%). Each training example is one NB walk ($T = 128$) on a CSL graph, and the class is predicted from mean-pooled features. At test time, each graph is judged from 16 walks ($T = 256$). As an *exact-count reference* we use logistic regression on the exact revisit frequencies at lags $1, \dots, S$. It is not an upper bound. At $S = 2$ it is at chance, because NB walks cannot return at lags 1–2, while the LTI model reaches 24.0%. The reference reaches 98.3%. For all 10 classes, its onset of 80% class accuracy is the first tested $S \geq W^*(r)$ (Table I). The trained models reach $60.4 \pm 7.9\%$ (LTI) and $55.2 \pm 8.5\%$ (input-dependent step) at $S = 64$. The largest tested S gives the best accuracy for both, and the curves are still rising. Neither remedy we tried

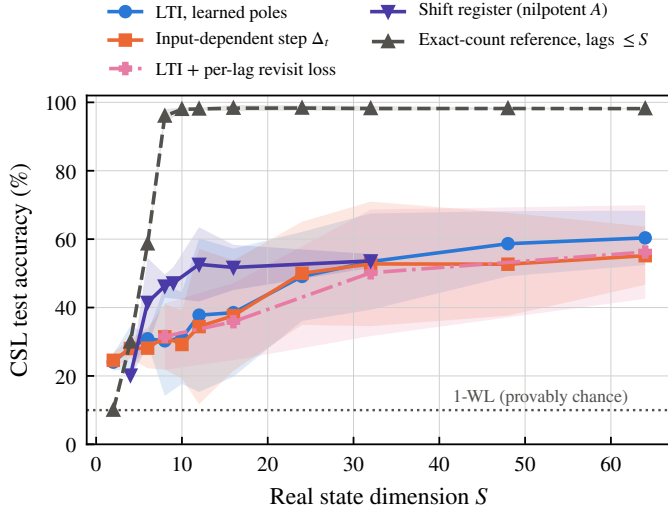


Fig. 4. CSL test accuracy against S (mean and 95% interval over 3 seeds; test graphs are the ten CSL graphs with fresh walks and features). The exact-count reference uses revisit frequencies at lags $\leq S$, which by Proposition 1(c) corresponds to an $(S+1)$ -state register.

TABLE I

CSL HORIZONS $W^*(r)$ AND PER-CLASS ONSETS: THE SMALLEST TESTED S WITH CLASS ACCURACY $\geq 80\%$ (MEAN OF 3 SEEDS; —: NOT REACHED FOR $S \leq 64$).

r	2	3	4	5	6	9	11	12	13	16
$W^*(r)$	3	4	5	6	7	8	7	8	5	7
Exact counts	4	4	6	6	8	8	8	8	6	8
LTI	2	6	—	—	—	—	—	—	32	—
Input-dep. Δ_t	2	8	—	—	—	—	—	—	64	—
Shift reg.	4	6	—	—	—	—	—	—	6	—

closes the gap:

- A per-lag revisit loss gives $56.2 \pm 13.7\%$ at $S = 64$.
- Evaluating at the training length ($T = 128$) instead of $T = 256$ lowers accuracy by 2.5 points.

The input-dependent step is -1.7 ± 1.1 points from LTI (paired over S and seeds). The dynamics are not the bottleneck either. A trained shift register has exact access to all lags $\leq S - 1$, with the same readout, pooling and labels. It reaches only 47.2% at $S = 9$, where the reference with lags ≤ 8 reaches 96.0%, and $53.6 \pm 2.0\%$ at $S = 32$. On CSL the gap therefore lies downstream of the dynamics, in the readout and mean pooling. Graph-level labels alone do not explain it, since the LTI model with a token-level revisit loss does no better. The shift register still tests Proposition 1(c) at the graph level: it first reaches 80% on classes $r = 2, 3$ and 13 at $S = 4, 6$ and 6 (Table I). This is the first tested $S \geq W^*(r) + 1$, the smallest register that holds lag $W^*(r)$. Class $r = 2$ ($W^* = 3$) is learned at $S = 2$ only by pooling: with one walk its accuracy is 53.7%, and with 16 walks it is 86.3%.

V. CONCLUSION

The pure delay has k unit Hankel singular values. For an order- S LTI state this gives three limits: for Gaussian inputs, exact detection needs $S \geq k + 1$ and a singular state matrix; the

explained variance of the lagged sample is at most S/k ; and every readout has divergence at most $\frac{1}{2} \ln \frac{k}{k-S}$. Optimal fits and a shift register attain these limits. Trained diagonal SSMs respect them but need several times the minimal state. For $k \leq 12$, frozen optimal poles remove most of this overhead. On random-feature inputs, where the gate stays inactive, an input-dependent step does not change it. On CSL graphs, even exact lag access through a shift register leaves trained models far below exact counts, so there the bottleneck lies downstream of the dynamics, in the readout and pooling. Lower bounds for input-dependent dynamics and better-conditioned delay parameterisations are natural next steps.

ACKNOWLEDGMENT

The authors used Claude (Anthropic), a generative AI system, extensively in this work. It wrote the experiment and analysis code, ran the experiments and produced all figures and tables, drafted and revised the text of every section, including the statements and proofs in Section III, and carried out simulated pre-submission reviews whose findings led to corrections. All code, data and generated numbers are released with the article. The authors take full responsibility for its content.

REFERENCES

- [1] A. Gu, K. Goel, and C. Ré, “Efficiently modeling long sequences with structured state spaces,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.
- [2] A. Gu, A. Gupta, K. Goel, and C. Ré, “On the parameterization and initialization of diagonal state space models,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2022.
- [3] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” in *Proc. Conf. Lang. Model. (COLM)*, 2024.
- [4] A. R. Voelker, I. Kajić, and C. Eliasmith, “Legendre memory units: Continuous-time representation in recurrent neural networks,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2019.
- [5] A. Gu, T. Dao, S. Ermon, A. Rudra, and C. Ré, “HiPPO: Recurrent memory with optimal polynomial projections,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2020.
- [6] B. L. Ho and R. E. Kalman, “Effective construction of linear state-variable models from input/output functions,” *Regelungstechnik*, vol. 14, no. 12, pp. 545–548, 1966.
- [7] K. Glover, “All optimal Hankel-norm approximations of linear multi-variable systems and their L^∞ -error bounds,” *Int. J. Control*, vol. 39, no. 6, pp. 1115–1193, 1984.
- [8] V. M. Adamjan, D. Z. Arov, and M. G. Kreĭn, “Analytic properties of Schmidt pairs for a Hankel operator and the generalized Schur–Takagi problem,” *Math. USSR-Sbornik*, vol. 15, no. 1, pp. 31–73, 1971.
- [9] Z. Li, J. Han, W. E, and Q. Li, “On the curse of memory in recurrent neural networks: Approximation and optimization analysis,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.
- [10] S. Jelassi, D. Brandfonbrener, S. M. Kakade, and E. Malach, “Repeat after me: Transformers are better than state space models at copying,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2024.
- [11] S. Arora, S. Eyuboglu, A. Timalina, I. Johnson, M. Poli, J. Zou, A. Rudra, and C. Ré, “Zoology: Measuring and improving recall in efficient language models,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.
- [12] W. Merrill, J. Petty, and A. Sabharwal, “The illusion of state in state-space models,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2024.
- [13] K. Xu, W. Hu, J. Leskovec, and S. Jegelka, “How powerful are graph neural networks?” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2019.
- [14] C. Morris, M. Ritzert, M. Fey, W. L. Hamilton, J. E. Lenssen, G. Rattan, and M. Grohe, “Weisfeiler and Leman go neural: Higher-order graph neural networks,” in *Proc. AAAI Conf. Artif. Intell.*, 2019.
- [15] R. L. Murphy, B. Srinivasan, V. Rao, and B. Ribeiro, “Relational pooling for graph representations,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2019.

- [16] V. P. Dwivedi, C. K. Joshi, A. T. Luu, T. Laurent, Y. Bengio, and X. Bresson, "Benchmarking graph neural networks," *J. Mach. Learn. Res.*, vol. 24, no. 43, pp. 1–48, 2023.
- [17] J. Tönshoff, M. Ritzert, H. Wolf, and M. Grohe, "Walking out of the Weisfeiler Leman hierarchy: Graph learning beyond message passing," *Trans. Mach. Learn. Res.*, 2023.
- [18] D. Chen, T. H. Schulz, and K. Borgwardt, "Learning long range dependencies on graphs via random walks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.
- [19] J. Kim, O. Zaghen, A. Suleymanzade, Y. Ryou, and S. Hong, "Revisiting random walks for learning on graphs," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.
- [20] R. Abboud, İ. İ. Ceylan, M. Grohe, and T. Lukasiewicz, "The surprising power of graph neural networks with random node initialization," in *Proc. Int. Joint Conf. Artif. Intell. (IJCAI)*, 2021.
- [21] R. Sato, M. Yamada, and H. Kashima, "Random features strengthen graph neural networks," in *Proc. SIAM Int. Conf. Data Mining (SDM)*, 2021.
- [22] C. Wang, O. Tsepa, J. Ma, and B. Wang, "Graph-Mamba: Towards long-range graph sequence modeling with selective state spaces," *arXiv preprint arXiv:2402.00789*, 2024.
- [23] A. Behrouz and F. Hashemi, "Graph Mamba: Towards learning on graphs with state space models," in *Proc. ACM SIGKDD Conf. Knowl. Discov. Data Mining (KDD)*, 2024.
- [24] K.-i. Hashimoto, "Zeta functions of finite graphs and representations of p -adic groups," in *Automorphic Forms and Geometry of Arithmetic Varieties*, ser. Adv. Stud. Pure Math. Academic Press, 1989, vol. 15, pp. 211–280.
- [25] N. Alon, I. Benjamini, E. Lubetzky, and S. Sodin, "Non-backtracking random walks mix faster," *Commun. Contemp. Math.*, vol. 9, no. 4, pp. 585–603, 2007.
- [26] C. Eckart and G. Young, "The approximation of one matrix by another of lower rank," *Psychometrika*, vol. 1, no. 3, pp. 211–218, 1936.
- [27] L. Mirsky, "Symmetric gauge functions and unitarily invariant norms," *Quart. J. Math.*, vol. 11, no. 1, pp. 50–59, 1960.
- [28] Z. Nehari, "On bounded bilinear forms," *Ann. Math.*, vol. 65, no. 1, pp. 153–162, 1957.